

Building a “Perfect EV Fit” Questionnaire for OFFOLabs

Executive summary

Your current intake (primary charging location, weekly miles, climate by ZIP, and “long-day” frequency) captures the **core EV constraints**—charging access and driving demand—but it cannot reliably produce a “perfect fit” because it misses the **biggest real-world drivers of EV satisfaction**:

- **Charging feasibility details** (home electrical capacity, dwelling type/permissions, dwell time, public charging reliability/price tolerance). Home charging is often sufficient for many drivers (sometimes with Level 1), but when it isn't, the difference between Level 1 and Level 2 infrastructure matters—and it depends on battery size, schedule, and dwell time. ¹
- **Range buffer needs** that change with cold weather, HVAC use, and driving style. AAA testing shows large range impacts in cold with HVAC, which your tool currently approximates but doesn't quantify or personalize. ²
- **Trip patterns** (where long trips go, whether destination charging exists, whether towing/roof racks/cargo loads are common). Load/towing can reduce range substantially in some contexts and should be asked explicitly when recommending body style and drivetrain. ³
- **Budget and ownership structure** (price ceiling, lease vs buy, incentives eligibility). Cost remains a primary barrier in multiple surveys and reports; without budget you can't choose an appropriate segment or avoid recommending mismatched trims. ⁴
- **Preference and lifestyle fit** (body style, passenger/cargo needs, hobbies, tech preferences, service access), which often determines satisfaction more than raw range.

A “perfect fit” tool should therefore evolve into a **two-layer system**:

- 1) a **constraint solver** (charging + range + budget + utility) that filters out bad options; and
- 2) a **preference scorer** (comfort, tech, UX, brand/service, lifestyle) that ranks the remaining “good fits.”

This report provides a research-grounded question set, a recommended scoring/logic model, and implementation guidance (including privacy cues and optional integrations with authoritative EV data sources from U.S. Environmental Protection Agency ⁵ and National Highway Traffic Safety Administration ⁶ for range ratings, VIN decode, and recalls). ⁷

Why the current questionnaire cannot produce a “perfect fit”

Charging “where” is not enough without charging “capability”

Your first question (home/work/public) identifies the dominant charging context, but it misses **charging capability**, which is what actually determines friction:

- Home charging feasibility depends on basics like electrical capacity and installation constraints. U.S. Environmental Protection Agency ⁵ notes that a home with ~200A service and breaker space “likely” has capacity for Level 2 charging, and that even 100A can often be sufficient depending on loads—meaning feasibility varies and should be assessed. ⁸
- U.S. Department of Energy ⁹ ’s Alternative Fuels Data Center states many EV owners can meet daily needs with Level 1 overnight charging if a dedicated outlet is available, but Level 2 becomes important for longer commutes, irregular schedules, or larger batteries. ¹⁰
- Dwell time (“how long the EV is at one location and can charge”) is explicitly called out as a determinant of whether Level 1 or Level 2 is needed. ¹¹

Implication: You need questions that distinguish *access* (can you charge at home) from *throughput* (how many miles of range do you reliably add overnight).

Weekly miles alone doesn’t model the “hard day”

Range satisfaction isn’t driven by average weekly miles; it’s driven by the hardest day and the buffer a user needs. Cold/heat and HVAC use also matter:

- AAA ¹² ’s range testing report found HVAC use at 20°F decreased combined range by ~41% compared to 75°F testing conditions, and hot conditions also decreased efficiency/range. ¹³

Implication: Ask about the pattern of “hard days,” not just frequency, and ask about typical highway speeds, parking outside, and whether preconditioning is used (or can be used).

Public charging trust and reliability are behavioral variables

If a user is public-dependent, their experience depends on willingness to tolerate variability in price/reliability and to plan charging:

- A Harvard Business School research summary characterizes charging pricing models as a “Wild West” pain point for EV drivers and dissatisfaction with charging pricing. ¹⁴
- An NBER working paper highlights that EV adoption depends on beliefs about charger reliability and that many Americans doubt sufficient buildout; it further uses real-time charging data to study adoption dynamics. ¹⁵
- Research on “charging anxiety” describes it as a minimum state-of-charge threshold where users become prohibitively anxious; this is a measurable behavioral parameter you can ask directly. ¹⁶

Implication: Add questions that quantify *planning tolerance* and *minimum comfortable battery level*.

“Perfect fit” requires budget, utility, and preference matching

Surveys and reports consistently show affordability and charging logistics as major adoption barriers:

- S&P Global reports cost and charging time/availability remain key barriers, and US consumers cite insufficient public charging in their area. ¹⁷
- AAA survey reporting includes concerns about residential charger installation and cost concerns. ¹⁸
- International Energy Agency ¹⁹ notes EV total cost of ownership can be favorable, but purchase price remains a barrier and affordability is central for broader uptake. ²⁰

Implication: You must ask budget constraints early and continuously: price cap, monthly payment target, lease vs buy, and incentive sensitivity.

Expanded EV Fit question framework

A practical approach is **progressive disclosure**: a quick “core fit” pass (2–4 minutes) followed by a deeper “perfect fit” pass (8–12 minutes) that branches only where needed.

Question modules and what each enables

The table below proposes additional questions, why they matter (research rationale), and how they change recommendations.

Module	High-leverage questions to add	Why it matters (research-grounded rationale)	What it enables in your EV fit engine
Charging reality	What type of home do you live in? (single-family / apartment / condo / other). Do you own or rent? Can you install a charger? (yes/no/need permission). Closest outlet to parking? Typical overnight dwell time?	Home vs non-home is not binary; installation feasibility and dwell time drive whether Level 1 or Level 2 works. ²¹	Classifies user into “home-reliable,” “home-limited,” “public-dependent,” etc., and sets charging friction baseline
Home electrical capacity	Do you know your panel service? (60/100/150/200A/unknown). Any spare breaker slots? Any large electric loads? (electric oven, HVAC, dryer). Willing to install load management?	Electrical capacity is a common constraint for Level 2; EPA notes typical service thresholds and that 100A may suffice depending on loads. ²²	Determines “Level 2 feasible now” vs “Level 1 + strategy” vs “upgrade needed,” and affects EV recommendation aggressiveness

Module	High-leverage questions to add	Why it matters (research-grounded rationale)	What it enables in your EV fit engine
Public charging dependence	How often will you rely on public DC fast charging? Do you have a preferred network nearby? Tolerance for 20–40 minute stops? How annoyed are you by variable pricing?	Dissatisfaction with charging pricing is documented; reliability beliefs influence adoption. ²³	Controls ranking tradeoffs: fast-charge performance, route planning tools, buffer requirements
Driving pattern (core)	Replace “weekly miles” with: Typical weekday miles, typical weekend miles, longest day in the last month, longest day in a typical year, % highway, typical highway speed band	Hard-day + highway share better approximates range needs than weekly averages; cold/hot impacts compound with highway and HVAC loads. ²⁴	Computes a personalized minimum EPA-range requirement + buffer for weather/speed
Climate and storage	Do you park indoors or outdoors overnight? Do you regularly face sub-freezing temperatures? Do you want/need heat pump? Do you precondition while plugged in?	Cold weather + HVAC materially reduces range; winter charging friction is documented. ²⁵	Increases recommended range buffer; changes prioritization for thermal features
Payload, towing, and hobbies	Do you tow? (how often, weight range). Roof box/ bikes/skis? Frequent full-passenger loads?	Load and towing can significantly reduce EV range (e.g., AAA study on heavy cargo; towing articles document reductions). ³	Determines whether user needs a larger pack, different body style, or PHEV suggestion
Body style and space	Must-have body style (sedan/ hatch/crossover/SUV/truck/ van). Seats needed. Cargo needs (stroller, pets). Ground clearance / AWD needs	“Perfect fit” is often body/utility first; also impacts efficiency and range.	Filters model set; sets utility + comfort ranking
Budget and deal structure	Total price ceiling and monthly payment target. Lease vs buy. New vs used. Willing to buy out-of-state.	Price is consistently a top barrier; the tool must respect budget early. ²⁶	Determines feasible segment, trims, and whether used/lease strategies should appear

Module	High-leverage questions to add	Why it matters (research-grounded rationale)	What it enables in your EV fit engine
Incentives and TCO sensitivity	Do you qualify for federal/state incentives? (unknown / yes / no). How important is cost-per-mile? Electricity rate awareness (enter \$/kWh or “unknown”)	EPA’s EV guidance highlights tax credits and refers users to incentives resources; electricity price drives operating costs. ²⁷	Enables a “cost-per-mile” view and a “best value EV” filter
Tech and UX priorities	Apple CarPlay / Android Auto requirement. Driver assist priority (basic/ advanced). App-first vs physical buttons preference. Software update comfort	User satisfaction often hinges on UX and driver-assist features rather than specs.	Preference scoring and “good-to-have” ranking
Service, reliability, and support	Distance willing to travel for service. Preference for large service network. Comfort with mobile service / new brands	Service access shapes ownership satisfaction, especially for newer brands.	Adds friction score for certain brands depending on local footprints
Risk tolerance and planning style	Minimum comfortable battery %. Are you OK arriving at 10–15% SOC? Do you plan trips in advance or drive spontaneously?	“Charging anxiety” can be modeled through minimum SOC thresholds; complex adoption depends on reinforcement and comfort. ²⁸	Turns anxiety/planning into explicit buffer multipliers

A “cleaned” version of your existing questions

Your current questions are good—but they can be made more diagnostic:

- **Charging location (keep)**, but add **time parked there** (dwell time) and **charging power access** (Level 1 outlet vs Level 2 vs DC fast). Dwell time and Level differences matter. ²⁹
- **Weekly miles (upgrade)** into day-pattern + longest day (see above).
- **Climate (keep)** but add *parking exposure* and *winter behavior*; cold has large measurable impacts. ³⁰
- **Longer-than-usual days (keep)** but ask *where* and *whether destination charging exists*, because that determines whether “faster charging” or “more range” is the correct solution.

Fit logic and outputs: how to turn answers into “perfect EV” recommendations

A robust fit engine should produce **three outputs**, not one list:

- **Vehicle shortlist** (ranked with transparent reasons)
- **Charging plan** (what to install/do; where to charge weekly)
- **Ownership plan** (budget approach, lease vs buy cues, incentive prompts)

A research-grounded scoring model

A practical model for OFFO is a two-stage approach:

Constraint filtering

Exclude vehicles that fail hard constraints:

- **Range constraint:** required minimum EPA range $\geq \text{LongestDayMiles} \times (1 + \text{buffer})$
- Buffer is a function of cold/heat, outdoor parking, highway share, and anxiety threshold. Cold impacts are documented and can justify larger buffers. ³¹
- **Charging constraint:** if user is public-dependent and dislikes unpredictability or long waits, prioritize vehicles known for stronger fast-charge usability and/or recommend PHEV option (if allowed). Public charging dissatisfaction and reliability beliefs matter. ³²
- **Budget constraint:** MSRP/used price range and financing/lease constraints.

Preference scoring

Score remaining vehicles on weighted dimensions:

- **Charging Fit Score** (home feasibility + commute + public reliance)
- **Range/Trip Fit Score** (hard day + buffer + road-trip frequency)
- **Utility Fit Score** (body style, cargo, towing) with explicit penalties for towing-heavy users due to documented range drops in load scenarios. ³³
- **Cost Fit Score** (price + operating cost estimates)
- **Tech Fit Score** (infotainment/driver assist preferences)
- **Ownership Friction Score** (service distance, comfort with new brands)

Add a “charging behavior persona” to personalize recommendations

A useful research-backed shortcut is to cluster users into behavior types, similar to how a 2025 survey-based study of U.S. EV owners identified distinct charging behavior groups (e.g., “range-anxious” vs “infrequent public charging” users). ³⁴

OFFO can do this with 3–5 questions: - minimum comfortable SOC

- frequency of public fast charging
- planning tolerance
- home charging feasibility

Then tailor: - “Range-anxious + public-dependent” → recommend larger buffers and stronger charging support tools

- “Home-reliable + predictable commute” → can recommend efficient, lower-range vehicles confidently

Data sources, privacy cues, and transparency requirements

Authoritative data sources you can safely rely on

To improve accuracy and user trust, you can pair the questionnaire with authoritative data:

- **Official efficiency and range rating data:** U.S. Environmental Protection Agency ⁵ provides downloadable fuel economy datasets and explains testing/label methodology (and that manufacturers run tests with EPA oversight and EPA confirms a portion of results). ³⁵
- **Charging infrastructure availability:** Alternative Fuels Data Center (AFDC) ³⁶ provides station locator datasets and developer APIs via NREL for alternative fuel stations including EV charging. ³⁷
- **VIN decode / vehicle identity:** National Highway Traffic Safety Administration ⁶ vPIC API recommends sending model year for more accurate decoding, and the VIN decoder is publicly documented. ³⁸
- **Recall status:** NHTSA provides recall lookup tools and datasets/APIs. ³⁹

Critical privacy note: VIN and ZIP are sensitive

One NHTSA presentation explicitly states that a VIN is specific to a vehicle and is considered PII. ⁴⁰

That means OFFOLabs should treat VIN storage like sensitive-data handling:

- Explain **why** VIN is requested (vehicle matching, recall checks, building a “garage”).
- Explain **what is stored** (full VIN vs hashed, retention period).
- Offer a clear **delete/export** option for saved vehicles.
- For ZIP-based climate detection, explain whether you store ZIP, approximate location, or only derived climate category.

Transparency guardrails for trust

Users will trust “perfect fit” recommendations only if outcomes are predictable:

- If any “premium report” or paywall exists, surface it early and clearly—avoid surprise gates.
- For any claim of “independent rating,” show:
 - data inputs (EPA range, NHTSA recalls, charging density via AFDC, etc.),
 - timestamps (“data as of ...”), and
 - limitations (e.g., public charging availability changes; vehicle trims vary).

Implementation roadmap with experimentation

Start with two experiences: Quick Fit and Deep Fit

- **Quick Fit (2–4 minutes):** your existing 4 questions + budget cap + body style + home charging feasibility question.

- **Deep Fit (8-12 minutes):** adds towing/cargo, tech preferences, service tolerance, planning/anxiety, and electricity cost sensitivity.

This reduces drop-off while still enabling power users to get “perfect fit” quality.

High-priority A/B tests and success metrics

Experiment	Hypothesis	Primary metric	Secondary metrics	Expected window
Add “budget + body style” early	Early budget/body style reduces irrelevant recommendations and increases completion	Completion rate of questionnaire	Downstream click-through to vehicle details; “satisfied with recommendations” survey	2-3 weeks
Add “hard day” question (longest day)	“Hard day” predicts better fit than weekly miles	Recommendation satisfaction	Lower “range anxiety” complaints; fewer users changing range filters later	2-4 weeks
Add “minimum comfortable SOC”	Anxiety-threshold personalization improves trust	Save/shortlist rate	Return rate to dashboard; time-to-first shortlist	2-4 weeks
Home charging feasibility module	Better charging plan improves adoption	% who reach “Charging plan” page	Drop-off at data request steps; customer support contacts	2-4 weeks

What to measure to prove “perfect fit” is working

At minimum, instrument:

- questionnaire completion and step drop-offs
- “recommendation acceptance” (shortlist/save rate)
- follow-up intent (dealer message / test drive scheduling if applicable later)
- subjective satisfaction (“These options match my needs”)
- correction behavior (users who immediately tighten filters after seeing results)

Practical “next questions” you can copy into the product now

Below are the most high-impact additions if you want to expand immediately without redesigning everything.

Must-add (minimal lift, maximal impact) - Price ceiling (and/or monthly payment target)

- Body style preference

- Home charging feasibility: own/rent + install permission + overnight dwell time

- Longest driving day in the past month (miles) + longest day in a typical year
- Parking location overnight (garage/outside/street)
- Minimum comfortable battery % (e.g., “I’m fine arriving at 10–15%” vs “I want 30%+ buffer”)

Strong “perfect fit” adds - Towing/cargo/roof rack usage

- Road trip destinations and destination charging availability
- Tech essentials (CarPlay/Android Auto, driver assist)
- Service access tolerance

If you want, I can translate this into a final **progressive question flow** (including branching logic and the exact UI wording) and a **transparent scoring explanation** (“Why we recommended these EVs”) aligned to the authoritative data sources above.

1 10 21 Charging Electric Vehicles at Home - AFDC

https://afdc.energy.gov/fuels/electricity-charging-home?utm_source=chatgpt.com

2 13 19 24 25 30 31 AAA ELECTRIC VEHICLE RANGE TESTING

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